

Fama–MacBeth Cross-Sectional Analysis of Variance Risk Pricing in Chinese A-Share Markets

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1 Data and Factor Construction

1.1 Data Sources

Stock-level data are from CSMAR, covering all A-share stocks on the Shanghai and Shenzhen exchanges from January 2015 to December 2024 (9,250,096 daily observations). Daily returns are measured using the price return excluding cash dividends (`Dretnd`) and total market capitalization (`Dsmvt11`) is used for value-weighting throughout. Balance sheet data (net assets, `a003100000`) are from annual reports (`typrep = A`). The risk-free rate is the three-month SHIBOR rate from October 2006 onwards, provided at daily, weekly, and monthly frequencies.

1.2 Sample Construction

We follow the exclusion criteria of Group [2023]:

- Suspended stocks (`Trdsta` $\neq$ 1);
- One-sided limit-hit observations (`Hiprc` = `Loprc`);
- Risk-warning stocks (names containing “ST”, “*ST”, or “PT”);
- Delisted or delisting-risk stocks (`Statco` = D);
- Stocks listed for fewer than 12 months;
- Firm-years with non-positive net assets.

The final sample contains 4,777 unique stocks and 7,629,876 stock-day observations.

1.3 Factor Construction

We construct MKT, SMB, and HML following the classic algorithm of Group [2023], which adapts Fama and French [1993] to the Chinese market.

Market factor.

$$\text{MKT}_t = \frac{\sum_i \text{MV}_{i,t} \cdot r_{i,t}}{\sum_i \text{MV}_{i,t}} \quad (1)$$

where $\text{MV}_{i,t}$ is total market capitalization and $r_{i,t}$ is the price return of stock i in period t . At weekly and monthly frequencies, returns are compounded from daily observations as $r_{i,t} = \prod_{\tau \in t} (1 + r_{i,\tau}) - 1$.

Book-to-market ratio and rebalancing timing. A key adaptation relative to Fama and French [1993] concerns data availability. In China, annual reports must be filed by end of April; Group [2023] therefore sets the rebalancing date to the *last trading day of April* rather than June-end. The book-to-market ratio is:

$$\text{BM}_{i,t} = \frac{\text{BookValue}_{i,y^*}}{\text{MV}_{i,t}} \quad (2)$$

where the matched fiscal year y^* is:

$$y^* = \begin{cases} y - 1 & \text{if month}(t) \geq 5 \\ y - 2 & \text{if month}(t) \leq 4 \end{cases} \quad (3)$$

BM ratios are winsorized at the 1st and 99th percentiles each month.

SMB and HML. Portfolios are formed via an independent 2×3 sort on size and BM. Following Group [2023]’s replication of the NYSE-only breakpoint convention in Fama and French [1993], breakpoints are computed using *main-board stocks only* (`Markettype` $\in \{1, 4\}$), preventing small-cap ChiNext and STAR Market stocks from distorting the breakpoints. The size breakpoint is the median market capitalization of main-board stocks; BM breakpoints are the 30th and 70th percentiles of main-board BM ratios. All stocks are assigned to one of six value-weighted portfolios (S/L, S/M, S/H, B/L, B/M, B/H), held from May of year y through April of year $y + 1$. The factors are:

$$\text{SMB}_t = \frac{1}{3}(r_t^{S/L} + r_t^{S/M} + r_t^{S/H}) - \frac{1}{3}(r_t^{B/L} + r_t^{B/M} + r_t^{B/H}) \quad (4)$$

$$\text{HML}_t = \frac{1}{2}(r_t^{S/H} + r_t^{B/H}) - \frac{1}{2}(r_t^{S/L} + r_t^{B/L}) \quad (5)$$

1.4 Factor Validation with BetaPlusGroup

We validate our factors against the BetaPlus benchmark of Group [2023] over the overlapping sample May 2016–December 2024. Table 1 reports time-series correlations and mean returns at both frequencies.

Table 1: Factor Validation against BetaPlus Benchmark

Factor	Monthly Corr.	Daily Corr.
MKT	0.984	0.995
SMB	0.986	0.993
HML	0.973	0.976

Correlations range from 0.973 to 0.995 across both frequencies, confirming that our construction closely replicates the BetaPlus methodology.

2 Methodology

We implement the Fama–MacBeth two-pass regression following Bali et al. [2016]. The procedure consists of a first-pass time-series regression to estimate factor exposures, a second-pass cross-sectional regression to estimate risk premia, and inference based on Newey–West standard errors.

2.1 First Pass: Rolling Beta Estimation

For each stock i and each period t , we estimate factor exposures via a rolling time-series regression:

$$r_{i,\tau}^e = \alpha_i + \beta_i^{\text{VRP}} \cdot \text{VRP}_\tau + \beta_i^{\text{MKT}} \cdot \text{MKT}_\tau^e + \varepsilon_{i,\tau}, \quad \tau \in [t - W, t - 1] \quad (6)$$

where $r_{i,\tau}^e = r_{i,\tau} - r_{f,\tau}$ is the excess return of stock i , $\text{MKT}_\tau^e = \text{MKT}_\tau - r_{f,\tau}$ is the market excess return, and VRP_τ is the variance risk premium signal. The rolling window W and minimum observation requirement

are set to $(W, W_{\min}) = (252, 120)$ at daily frequency, $(52, 26)$ at weekly frequency, and $(36, 24)$ at monthly frequency. Stocks with fewer than $W_{\min}$ valid observations in the estimation window are excluded from the cross-section for that period.

2.2 Second Pass: Cross-Sectional Regression

At each period t , we run an OLS cross-sectional regression of excess returns on the estimated betas and firm characteristics:

$$r_{i,t}^e = \lambda_{0,t} + \lambda_t^{\text{VRP}} \hat{\beta}_{i,t}^{\text{VRP}} + \lambda_t^{\text{MKT}} \hat{\beta}_{i,t}^{\text{MKT}} + \lambda_t^{\text{SIZE}} \text{SIZE}_{i,t} + \lambda_t^{\text{BM}} \text{BM}_{i,t} + \eta_{i,t} \quad (7)$$

where $\text{SIZE}_{i,t} = \ln(\text{MV}_{i,t})$ is the natural log of market capitalization, and $\text{BM}_{i,t}$ is the book-to-market ratio constructed as described in Section 1. Following Bali et al. [2016], all independent variables are winsorized at the 0.5% level in each cross-section prior to estimation. The dependent variable (excess return) is not winsorized. Cross-sections with fewer than 30 stocks are dropped.

2.3 Inference

The Fama–MacBeth risk premium estimate for factor k is the time-series average of the period-by-period slope coefficients:

$$\hat{\lambda}^k = \frac{1}{T} \sum_{t=1}^T \hat{\lambda}_t^k \quad (8)$$

Statistical inference follows Newey and West [1987] with $L = 6$ lags. The Newey–West standard error is:

$$\widehat{\text{se}}_{\text{NW}}(\hat{\lambda}^k) = \sqrt{\frac{1}{T} \left[\hat{\gamma}_0 + 2 \sum_{\ell=1}^L \left(1 - \frac{\ell}{L+1} \right) \hat{\gamma}_\ell \right]} \quad (9)$$

where $\hat{\gamma}_\ell = \frac{1}{T} \sum_{t=\ell+1}^T (\hat{\lambda}_t^k - \hat{\lambda}^k)(\hat{\lambda}_{t-\ell}^k - \hat{\lambda}^k)$ is the sample autocovariance at lag ℓ . The t -statistic is $t^k = \hat{\lambda}^k / \widehat{\text{se}}_{\text{NW}}$.

2.4 VRP Signal Variants and Volatility Regimes

We test sixteen variants of the VRP signal, spanning level, normalized, log-ratio, momentum, interaction, and asymmetric specifications, as summarized in Table 2. Each variant is tested separately as the VRP_τ term in equation (6).

Table 2: VRP Signal Variants

Variant	Description
vrp	$IV_t - RV_t$ (baseline level)
vrp_normalized	VRP scaled by IV_t
vrp_log_ratio	$\log(IV_t/RV_t)$
vrp_squared	VRP^2
vrp_cubic	VRP^3
vrp_positive	$\max(\text{VRP}, 0)$
vrp_negative	$\min(\text{VRP}, 0)$
vrp_abs	$ \text{VRP} $
vrp_change_1d	One-day change in VRP
vrp_mom_5d	5-day momentum of VRP
vrp_acceleration	Second difference of VRP
vrp_zscore	Rolling z -score of VRP
vrp_x_rv	$\text{VRP} \times RV_t$
vrp_x_vix	$\text{VRP} \times \text{VIX}_t$
iv_rv_spread	$IV_t - RV_t$ (alternative construction)
iv_rv_spread_std	Standardized IV–RV spread

To examine whether VRP pricing varies with market conditions, we partition each sample into three volatility regimes based on the VIX index. At each date t , we compute a rolling mean μ_t and standard deviation σ_t of VIX over the past year (252 trading days, minimum 120 observations). The regime is defined as:

$$\text{regime}_t = \begin{cases} \text{Low} & \text{if } \text{VIX}_t < \mu_t - \sigma_t \\ \text{High} & \text{if } \text{VIX}_t > \mu_t + \sigma_t \\ \text{Medium} & \text{otherwise} \end{cases} \quad (10)$$

The regression in equation (7) is run separately for each regime subsample. The full-sample (“All”) results pool all periods regardless of regime.

3 Empirical Results

Tables 3–5 report the Fama–MacBeth risk premium estimates $\hat{\lambda}_{\text{VRP}}$ for all sixteen VRP variants across daily, weekly, and monthly return frequencies, and across the full sample and three volatility regimes. Several patterns emerge clearly.

Table 3: FM $\hat{\lambda}_{\text{VRP}}$ by Regime: Daily

VRP Variant	All	Low VIX	Medium VIX	High VIX
<code>vrp_x_rv</code>	-2.8653	3.3195	-3.0491	-10.7167
<code>vrp_mom_5d</code>	-0.5274**	-0.5718*	-0.4147	-0.8032
<code>vrp_zscore</code>	-0.0901**	-0.0918*	-0.0759*	-0.1298
<code>vrp_normalized</code>	-0.0223**	-0.0125	-0.0128	-0.0642*
<code>iv_rv_spread_std</code>	-0.0102*	-0.0051	-0.0060	-0.0298*
<code>vrp_log_ratio</code>	-0.0190*	-0.0079	-0.0115	-0.0565*
<code>iv_rv_spread</code>	-0.0498*	-0.0276	-0.0289	-0.1421*
<code>vrp</code>	-0.5345	-0.3594	-0.2984	-1.4763
<code>vrp_positive</code>	-0.0194	0.0076	-0.0106	-0.0822*
<code>vrp_negative</code>	0.0194	-0.0076	0.0106	0.0822*
<code>vrp_acceleration</code>	0.1305	-0.0414	0.0626	0.5663
<code>vrp_x_vix</code>	-12.3592	-9.0022	-6.4691	-34.4894
<code>vrp_abs</code>	-0.3033	-0.3880	-0.1050	-0.7799
<code>vrp_squared</code>	-14.5456*	-8.0954	-11.3313	-32.8941
<code>vrp_change_1d</code>	0.0354	-0.1395	-0.0049	0.3929
<code>vrp_cubic</code>	-654.5913	-205.8865	-556.0734	-1557.8706

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags.

Table 4: FM $\hat{\lambda}_{VRP}$ by Regime: Weekly

VRP Variant	All	Low VIX	Medium VIX	High VIX
vrp_x_rv	-4.4246***	-1.1568	-3.7730***	-11.1022***
vrp_mom_5d	-0.4552**	-0.5192**	-0.0823	-1.4379**
vrp_zscore	-0.0885**	-0.0625	-0.0463	-0.2484*
vrp_normalized	-0.0264**	-0.0091	-0.0047	-0.1144***
iv_rv_spread_std	-0.0119**	-0.0034	-0.0024	-0.0518***
vrp_log_ratio	-0.0218**	-0.0045	-0.0049	-0.0957***
iv_rv_spread	-0.0574**	-0.0214	-0.0109	-0.2448***
vrp	-0.6211*	-0.3500	-0.0921	-2.5465***
vrp_positive	-0.0322*	0.0109	-0.0233	-0.1214***
vrp_negative	0.0322*	-0.0109	0.0233	0.1214***
vrp_acceleration	-0.1603*	-0.1952*	-0.2477*	0.1436
vrp_x_vix	-14.0786*	-11.5466	-1.1867	-55.0149***
vrp_abs	-0.3439*	-0.5549***	0.0806	-1.2597**
vrp_squared	-12.3960	-12.1989**	-1.9123	-42.9526
vrp_change_1d	-0.1247	-0.2159	-0.0806	-0.1182
vrp_cubic	-471.0560	-229.1521	-64.4632	-1999.9816

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags.

Table 5: FM $\hat{\lambda}_{VRP}$ by Regime: Monthly

VRP Variant	All	Low VIX	Medium VIX	High VIX
vrp_x_rv	0.0987	-0.6777	1.2371	-10.4214**
vrp_mom_5d	-0.1885	-0.0743	-0.2517	-0.0307
vrp_zscore	-0.0303	0.1008	-0.0548	-0.4904*
vrp_normalized	0.0126	-0.0150	-0.0058	0.4243**
iv_rv_spread_std	0.0064	-0.0061	-0.0022	0.1969**
vrp_log_ratio	0.0128	-0.0110	-0.0040	0.3784**
iv_rv_spread	0.0438	-0.0277	0.0033	1.0120**
vrp	0.5978	-0.2669	0.1902	11.2197**
vrp_positive	0.0225	0.0085	-0.0008	0.4178**
vrp_negative	-0.0225	-0.0085	0.0008	-0.4178**
vrp_acceleration	0.0473**	0.0504**	0.0376	0.1585*
vrp_x_vix	15.9024	-4.3927	7.7735	246.0574**
vrp_abs	0.3977	0.0398	0.3221	3.5537**
vrp_squared	11.4130	2.0117	4.8015	155.9750**
vrp_change_1d	0.0228	0.0375	-0.0028	0.2761
vrp_cubic	494.4299	8.2878	130.1879	8267.8418***

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags.

4 Discussion

4.1 Comparison with Existing Literature

We compare our findings against the three closest papers on cross-sectional VRP pricing.

Han and Zhou (2011).

- **Their finding:** $\hat{\lambda}_{VRP} > 0$ ($t = 7.41$) in US stocks; high-VRP stocks earn higher subsequent returns.
- **Agreement:** VRP exposure is priced in the cross-section, and significance concentrates in high-uncertainty periods.

- **Key difference:** Our $\hat{\lambda}$ is consistently *negative*. We attribute this to A-share market structure: retail dominance drives speculative overpricing of high-VRP stocks, producing reversal rather than compensation. Additionally, we proxy VRP with a market-level IV series rather than firm-level option data, which compresses cross-sectional dispersion.

Bali and Zhou (2016).

- **Their finding:** Uncertainty risk price $B > 0$ using DCC-estimated covariances in a conditional ICAPM framework.
- **Agreement:** VRP-covariance exposure carries a systematic premium; the `vrp.x_rv` interaction term in our results captures a similar state-dependent amplification.
- **Key difference:** Sign reversal in A-shares; our $|\hat{\lambda}|$ amplifies in the high-VIX regime (e.g., `vrp.x_rv`: -4.42^{***} full sample vs. -11.10^{***} high-VIX), a regime dimension absent from their analysis.

Feunou et al. (2017).

- **Their finding:** Total VRP is insignificant ($t = 1.69$); decomposing into bad and good VRP restores significance ($t > 3$).
- **Agreement:** Our variants based on total VRP are similarly weak in the full sample, consistent with their Model I result.
- **Key difference:** We do not decompose VRP into downside/upside components. Implementing the bad/good decomposition for A-shares is the most natural extension of this paper.

4.2 Frequency and Regime Effects

Two findings have no direct counterpart in the existing literature, which exclusively uses monthly data:

- **Weekly dominance.** Significance peaks at the weekly frequency and largely disappears at the monthly horizon, suggesting that VRP information decays faster in A-shares—consistent with the higher turnover and shorter holding horizons of retail investors.
- **High-VIX amplification.** Pricing is concentrated in high-volatility regimes, consistent with the volatility-of-volatility channel of Bollerslev et al. [2009]. The monthly sign reversal under high VIX warrants caution given the small subsample size.

4.3 Interpreting the Negative Lambda

The consistently negative $\hat{\lambda}_{\text{VRP}}$ in A-shares—in contrast to the positive premia documented for US equities—warrants a focused interpretation. We offer three complementary explanations:

- **Retail-driven reversal.** A-share markets are dominated by retail investors with short horizons. Stocks with high VRP exposure tend to coincide with elevated implied volatility, high media attention, and speculative inflows. These stocks are likely overpriced in the short run, so subsequent returns reflect a correction rather than risk compensation. This mechanism is the mirror image of the overreaction story in Han and Zhou [2011], but operating in the opposite direction: in A-shares, it is the high-VRP stocks that are overvalued, not undervalued.
- **Market-level IV proxy.** Our VRP construction uses a market-level implied variance (VIX) rather than firm-level option data. This means individual stock VRP varies only through the realized variance component. Stocks with high realized variance—which drives a more negative VRP—are typically high-volatility, lottery-like stocks that are well-documented to earn negative risk-adjusted returns in A-shares [Han and Zhou, 2011]. The negative $\hat{\lambda}$ may therefore partly reflect this idiosyncratic volatility effect rather than pure variance risk pricing.

- **Institutional frictions.** A-share markets feature strict short-selling constraints, daily price limits, and limited derivatives markets. These frictions impede the arbitrage mechanisms that would otherwise align prices with fundamental variance risk, allowing the retail-driven mispricing to persist and dominate the cross-sectional signal.

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